



# Customer Behaviour Analysis

## Import Libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("Ecommerce_customer_data.csv")
```

## Dataset Understanding

```
In [3]: df.head()
```

```
Out[3]:
```

|   | Customer ID | Purchase Date       | Product Category | Product Price | Quantity | Total Purchase Amount | Payment Method | Cus |
|---|-------------|---------------------|------------------|---------------|----------|-----------------------|----------------|-----|
| 0 | 46251       | 2020-09-08 09:38:32 | Electronics      | 12            | 3        | 740                   | Credit Card    |     |
| 1 | 46251       | 2022-03-05 12:56:35 | Home             | 468           | 4        | 2739                  | PayPal         |     |
| 2 | 46251       | 2022-05-23 18:18:01 | Home             | 288           | 2        | 3196                  | PayPal         |     |
| 3 | 46251       | 2020-11-12 13:13:29 | Clothing         | 196           | 1        | 3509                  | PayPal         |     |
| 4 | 13593       | 2020-11-27 17:55:11 | Home             | 449           | 1        | 3452                  | Credit Card    |     |

```
In [4]: df.shape
```

```
Out[4]: (250000, 13)
```

```
In [5]: df.columns
```

```
Out[5]: Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
              'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
              'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
              dtype='object')
```

```
In [6]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          250000 non-null  int64
1   Purchase Date                        250000 non-null  object
2   Product Category                     250000 non-null  object
3   Product Price                        250000 non-null  int64
4   Quantity                            250000 non-null  int64
5   Total Purchase Amount                250000 non-null  int64
6   Payment Method                       250000 non-null  object
7   Customer Age                         250000 non-null  int64
8   Returns                             202404 non-null  float64
9   Customer Name                        250000 non-null  object
10  Age                                  250000 non-null  int64
11  Gender                              250000 non-null  object
12  Churn                               250000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB

```

```
In [7]: df.dtypes
```

```

Out[7]: Customer ID          int64
Purchase Date          object
Product Category       object
Product Price          int64
Quantity               int64
Total Purchase Amount  int64
Payment Method         object
Customer Age           int64
Returns                float64
Customer Name          object
Age                    int64
Gender                 object
Churn                  int64
dtype: object

```

## Summary Statistics

```
In [8]: df.describe()
```

Out[8]:

|              | Customer ID  | Product Price | Quantity      | Total Purchase Amount | Customer Age |
|--------------|--------------|---------------|---------------|-----------------------|--------------|
| <b>count</b> | 250000.00000 | 250000.000000 | 250000.000000 | 250000.000000         | 250000.00000 |
| <b>mean</b>  | 25004.03624  | 254.659512    | 2.998896      | 2725.370732           | 43.94052     |
| <b>std</b>   | 14428.27959  | 141.568577    | 1.414694      | 1442.933565           | 15.35024     |
| <b>min</b>   | 1.00000      | 10.000000     | 1.000000      | 100.000000            | 18.00000     |
| <b>25%</b>   | 12497.75000  | 132.000000    | 2.000000      | 1477.000000           | 31.00000     |
| <b>50%</b>   | 25018.00000  | 255.000000    | 3.000000      | 2724.000000           | 44.00000     |
| <b>75%</b>   | 37506.00000  | 377.000000    | 4.000000      | 3974.000000           | 57.00000     |
| <b>max</b>   | 50000.00000  | 500.000000    | 5.000000      | 5350.000000           | 70.00000     |

In [9]: `df.median(numeric_only=True)`

```
Out[9]: Customer ID          25018.0
Product Price          255.0
Quantity                3.0
Total Purchase Amount  2724.0
Customer Age           44.0
Returns                0.0
Age                    44.0
Churn                  0.0
dtype: float64
```

In [10]: `df.mode().iloc[0]`

```
Out[10]: Customer ID          36437
Purchase Date      2022-05-08 12:58:55
Product Category      Clothing
Product Price        100.0
Quantity             1.0
Total Purchase Amount  2786.0
Payment Method      Credit Card
Customer Age         58.0
Returns              0.0
Customer Name      Michael Smith
Age                 58.0
Gender              Female
Churn               0.0
Name: 0, dtype: object
```

## Data Cleaning

In [11]: `df.isnull().sum()`

```
Out[11]: Customer ID          0
Purchase Date          0
Product Category       0
Product Price          0
Quantity              0
Total Purchase Amount  0
Payment Method         0
Customer Age           0
Returns                47596
Customer Name          0
Age                   0
Gender                0
Churn                 0
dtype: int64
```

```
In [12]: df.duplicated().sum()
```

```
Out[12]: 0
```

## Feature Engineering

```
In [13]: df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

```
In [14]: df['Month'] = df['Purchase Date'].dt.month_name()
```

```
In [15]: df['Year'] = df['Purchase Date'].dt.year
```

```
In [16]: df['Month_Year'] = df['Purchase Date'].dt.to_period('M')
```

```
In [17]: df['Return_Flag'] = df['Returns'].map({
    'Yes':1,
    'No':0
})
```

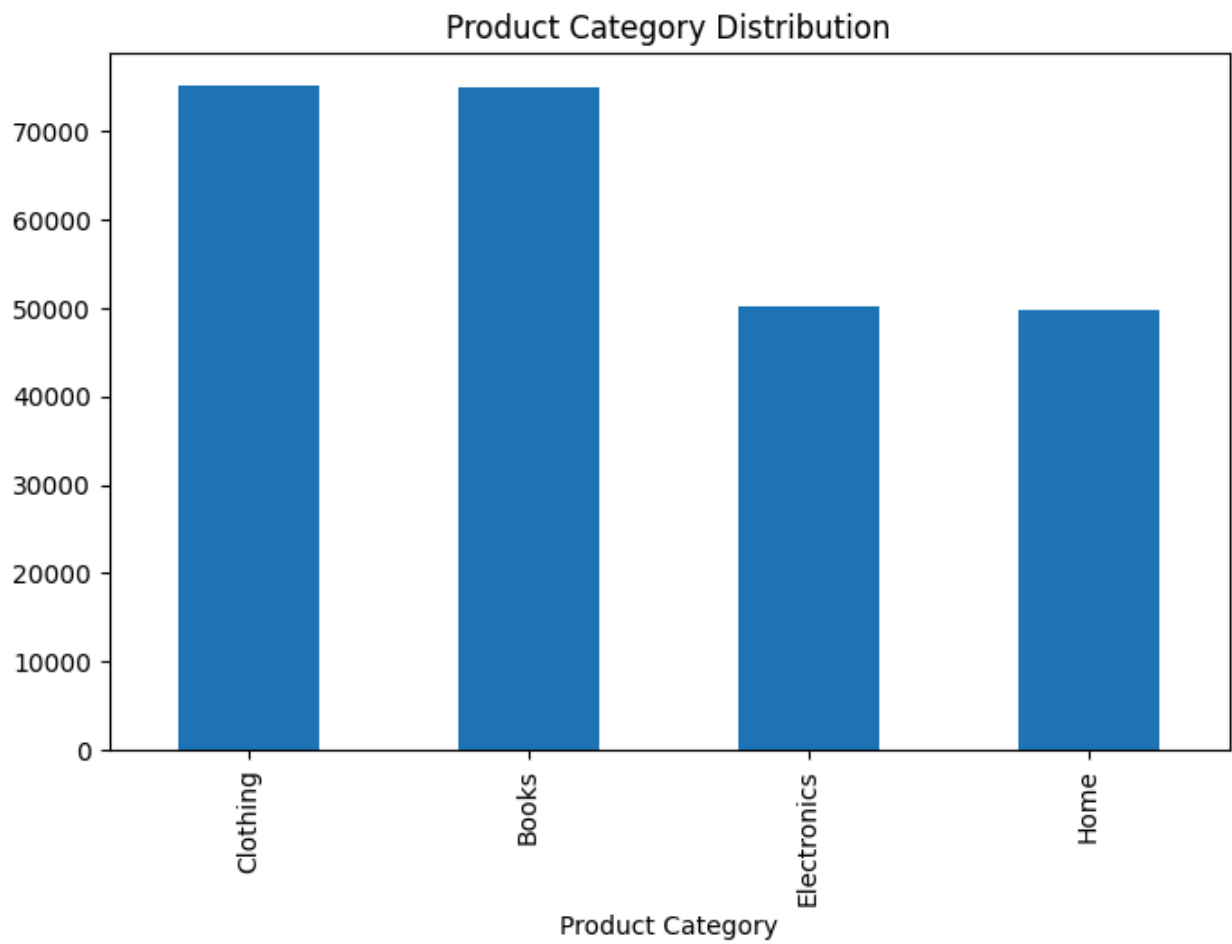
## Visualization

```
In [18]: #Product Category Distribution

plt.figure(figsize=(8,5))

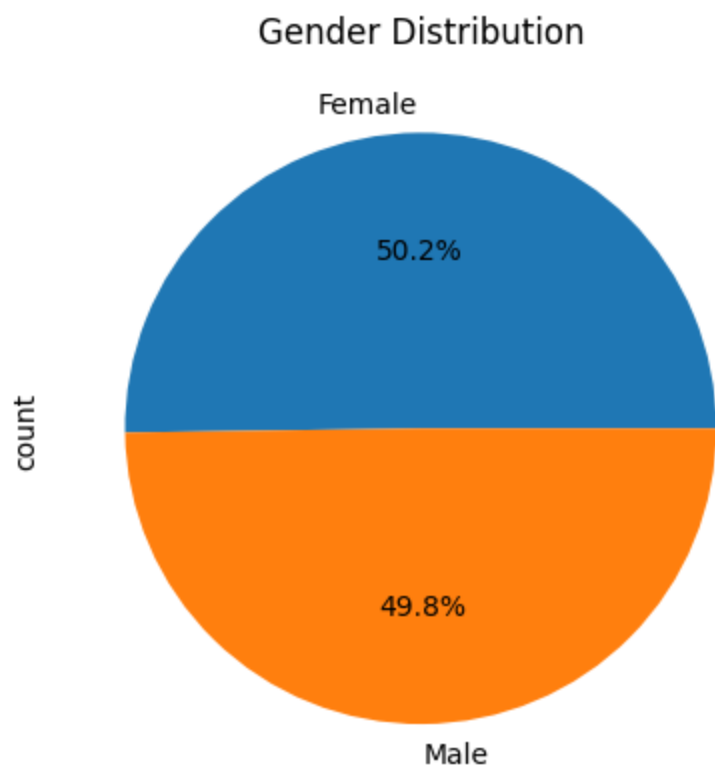
df['Product Category'].value_counts().plot(kind='bar')

plt.title("Product Category Distribution")
plt.savefig("product_category.png",dpi=300,bbox_inches="tight")
plt.show()
```



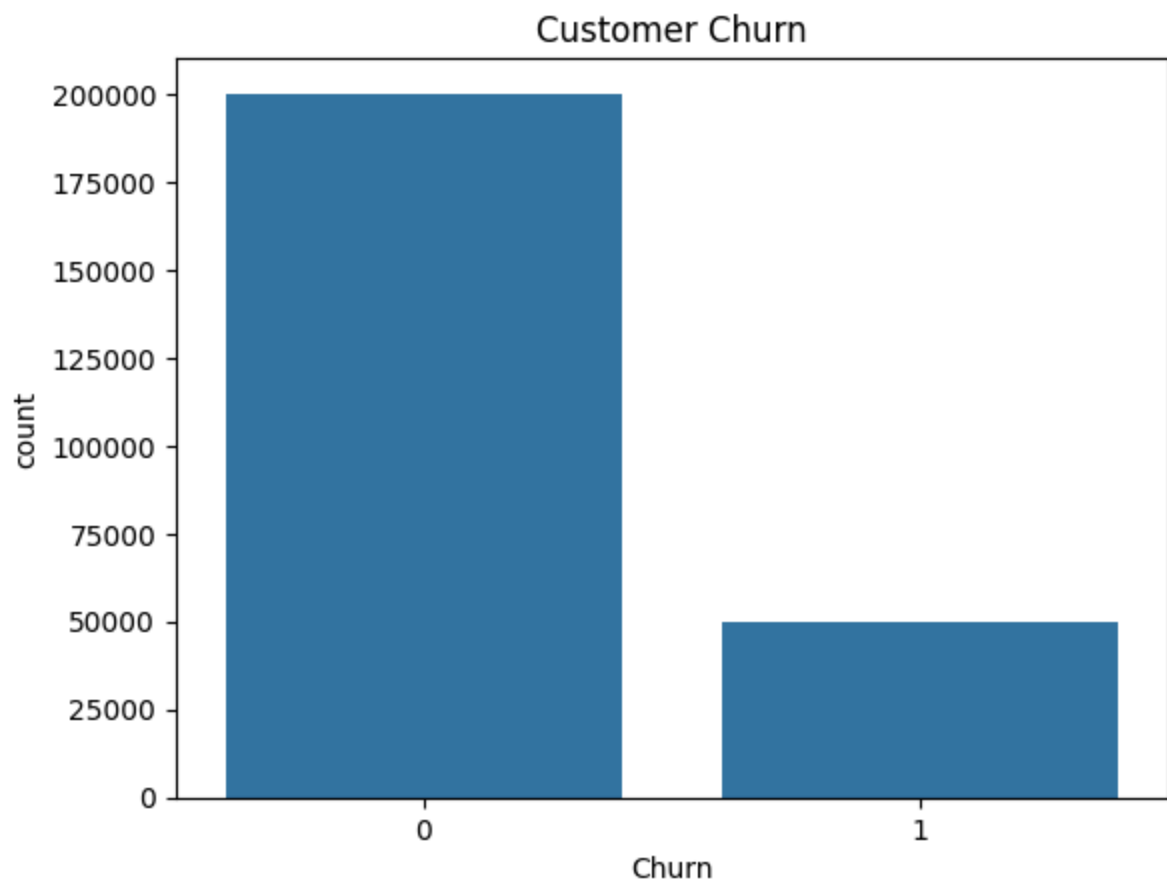
```
In [19]: #Gender Distribution
df['Gender'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Gender Distribution")
plt.savefig("gender_distribution.png",dpi=300,bbox_inches="tight")
plt.show()
```



In [20]: *#Churn distribution*

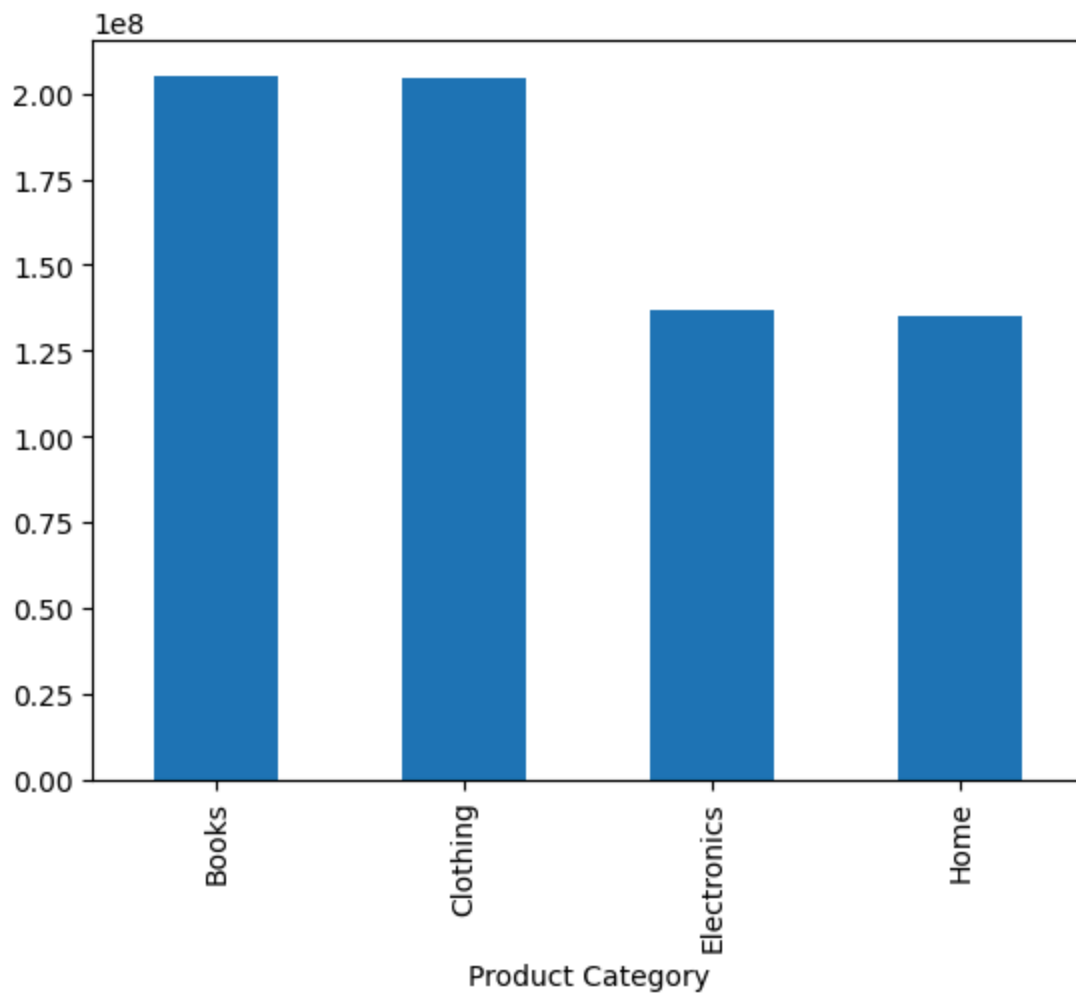
```
sns.countplot(  
    x='Churn',  
    data=df  
)  
  
plt.title("Customer Churn")  
plt.savefig("churn.png",dpi=300,bbox_inches="tight")  
plt.show()
```



```
In [21]: #Revenue by product category

revenue = df.groupby(
    'Product Category'
)['Total Purchase Amount'].sum()

revenue.sort_values(
    ascending=False
).plot(kind='bar')
plt.savefig("rev.png", dpi=300, bbox_inches="tight")
```

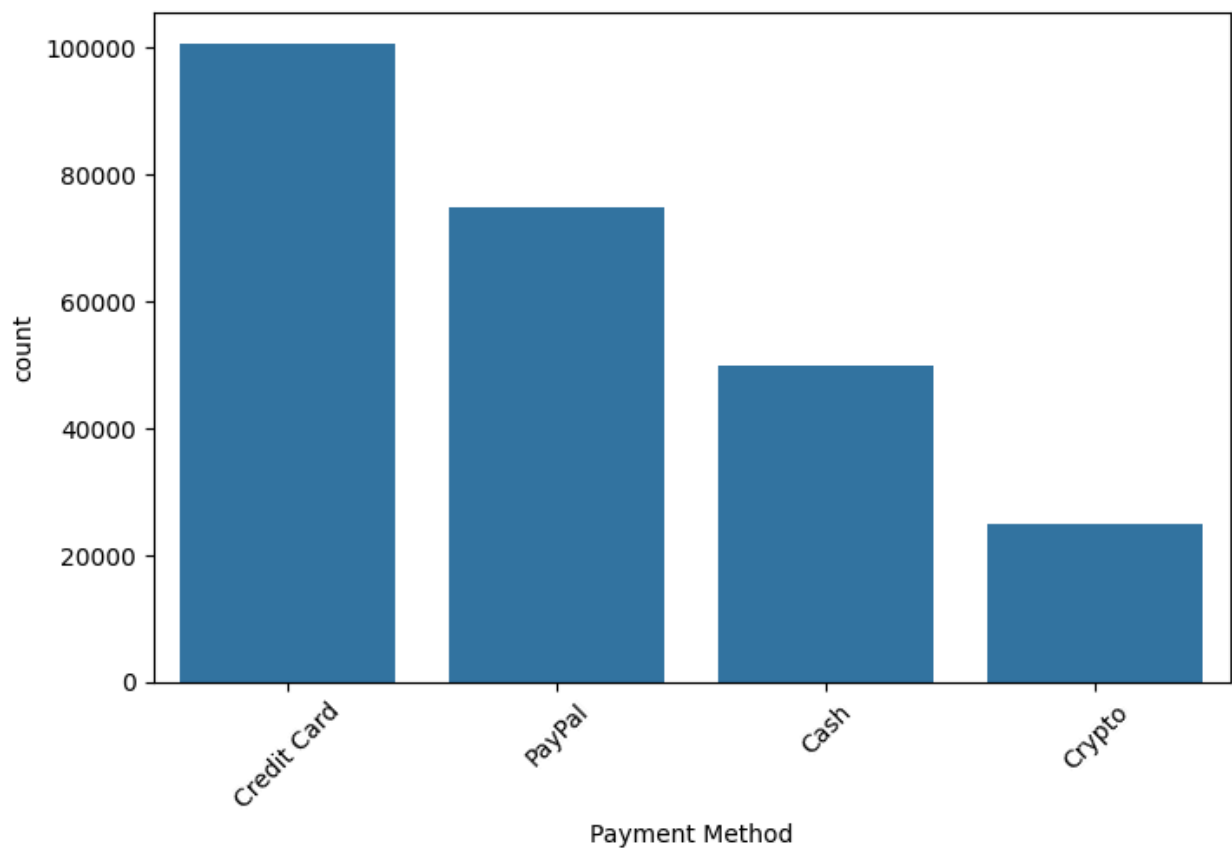


In [22]: *#payment method analysis*

```
plt.figure(figsize=(8,5))
sns.countplot(
    x='Payment Method',
    data=df
)

plt.xticks(rotation=45)
plt.show()
plt.savefig("payment.png",dpi=300,bbox_inches="tight")
```

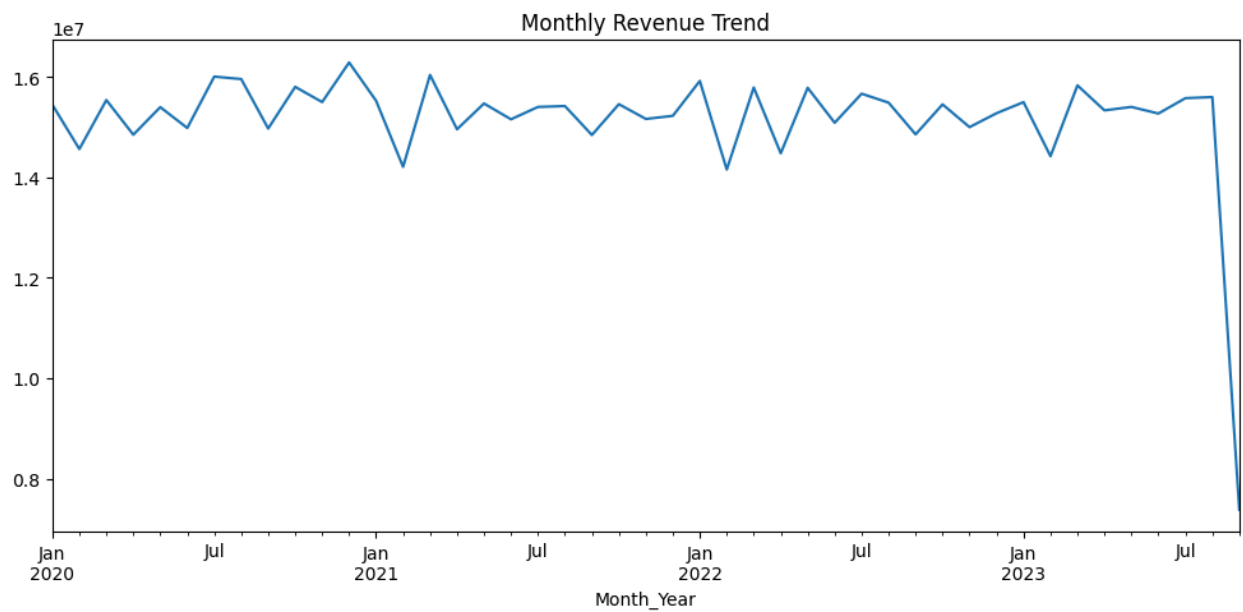




<Figure size 640x480 with 0 Axes>

In [23]: *#Monthly Sales Trend*

```
monthly_sales = df.groupby(  
    'Month_Year'  
)['Total Purchase Amount'].sum()  
  
monthly_sales.plot(  
    figsize=(12,5)  
)  
  
plt.title("Monthly Revenue Trend")  
plt.show()  
plt.savefig("monthly.png", dpi=300, bbox_inches="tight")
```



<Figure size 640x480 with 0 Axes>

In [24]: `#customer segmentation(RFM)`

```
snapshot_date = (
    df['Purchase Date'].max()
    + pd.Timedelta(days=1)
)
```

In [25]: `#Create RFM Table`

```
rfm = df.groupby('Customer ID').agg({

    'Purchase Date':
    lambda x:
    (snapshot_date - x.max()).days,

    'Customer ID': 'count',

    'Total Purchase Amount': 'sum'

})
```

In [26]: `#Rename Columns`

```
rfm.columns = [
    'Recency',
    'Frequency',
    'Monetary'
]
```

In [27]: `#RFM Scoring`

```
rfm['R'] = pd.qcut(
    rfm['Recency'],
    4,
    labels=[4,3,2,1])
```

```

)

rfm['F'] = pd.qcut(
    rfm['Frequency'].rank(method='first'),
    4,
    labels=[1,2,3,4]
)

rfm['M'] = pd.qcut(
    rfm['Monetary'],
    4,
    labels=[1,2,3,4]
)

```

```

In [28]: #Create Segments
rfm['Segment'] = 'Regular'

rfm.loc[
    (rfm['R']==4)&
    (rfm['F']==4)&
    (rfm['M']==4),
    'Segment'
] = 'Champions'

rfm.loc[
    (rfm['R']>=3)&
    (rfm['F']>=3),
    'Segment'
] = 'Loyal Customers'

rfm.loc[
    (rfm['R']==1),
    'Segment'
] = 'At Risk'

```

```

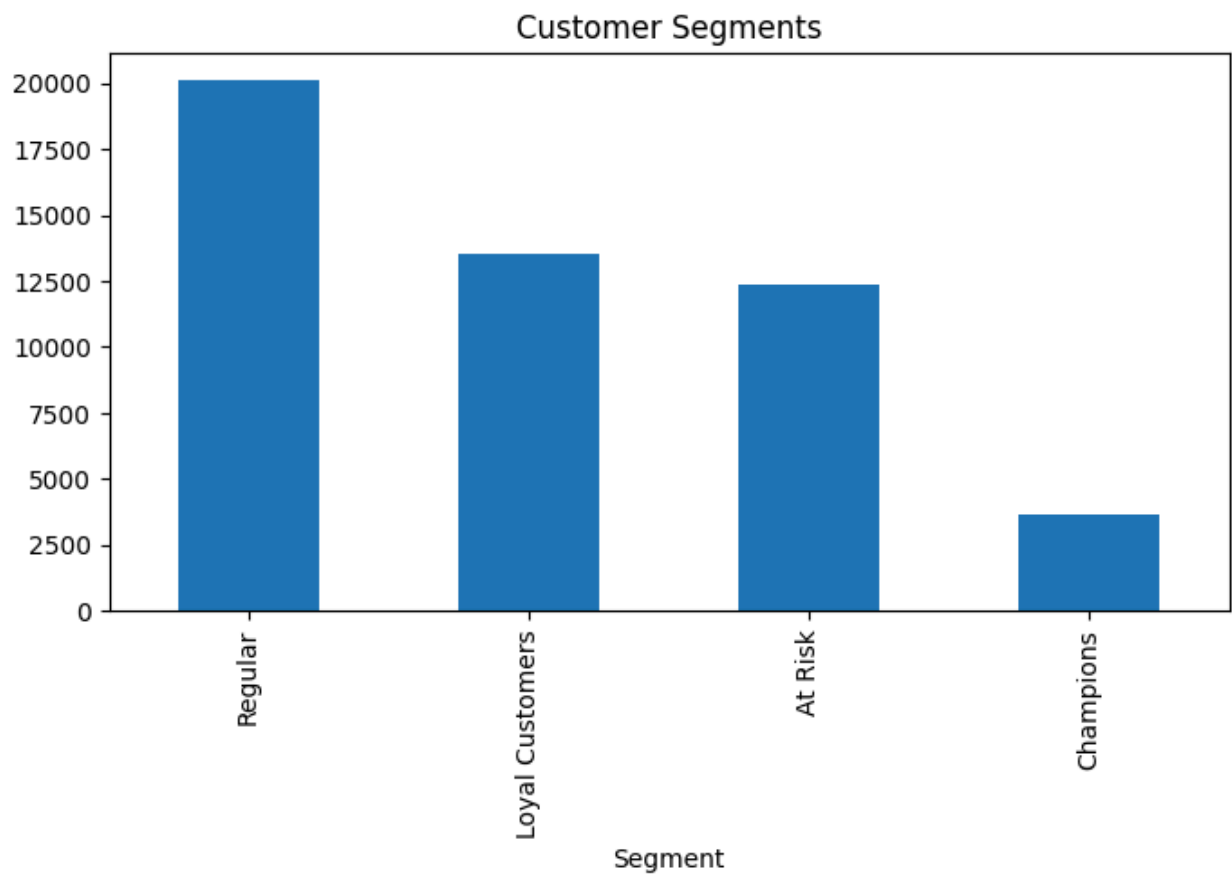
In [29]: #Segment Visualization

plt.figure(figsize=(8,4))
rfm['Segment'].value_counts().plot(
    kind='bar'
)

plt.title(
    "Customer Segments"
)

plt.show()
plt.savefig("segments.png",dpi=300,bbox_inches="tight")

```



<Figure size 640x480 with 0 Axes>

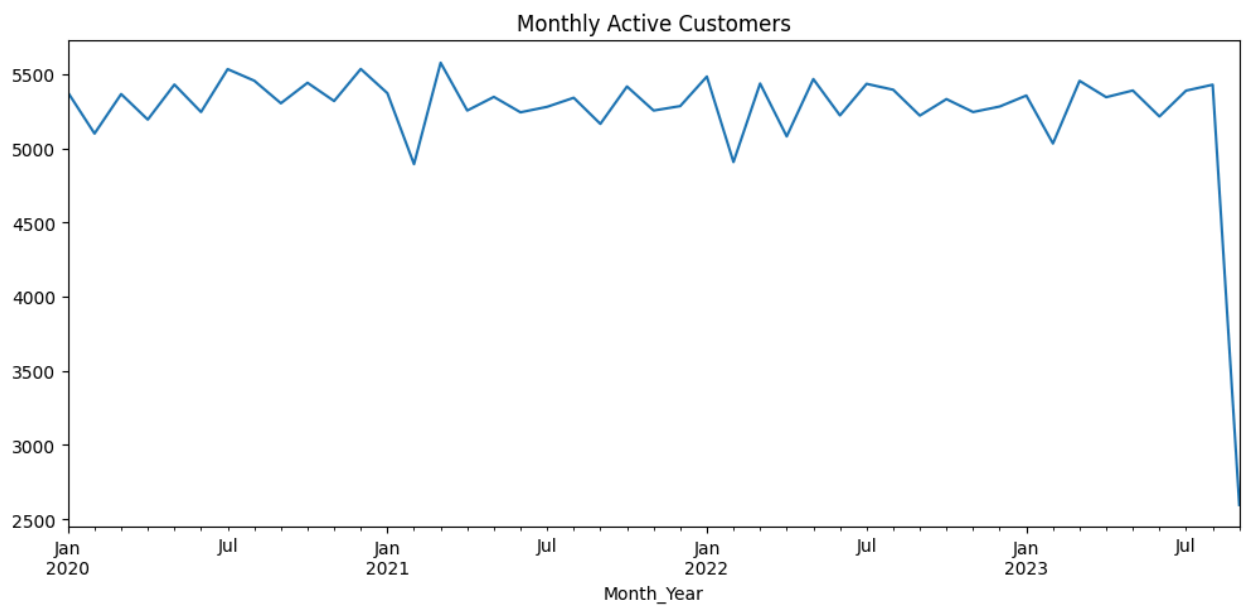
```
In [30]: #Retention Analysis

retention = df.groupby(
    'Month_Year'
)['Customer ID'].nunique()

retention.plot(
    figsize=(12,5)
)

plt.title(
    "Monthly Active Customers"
)

plt.show()
plt.savefig("active.png",dpi=300,bbox_inches="tight")
```

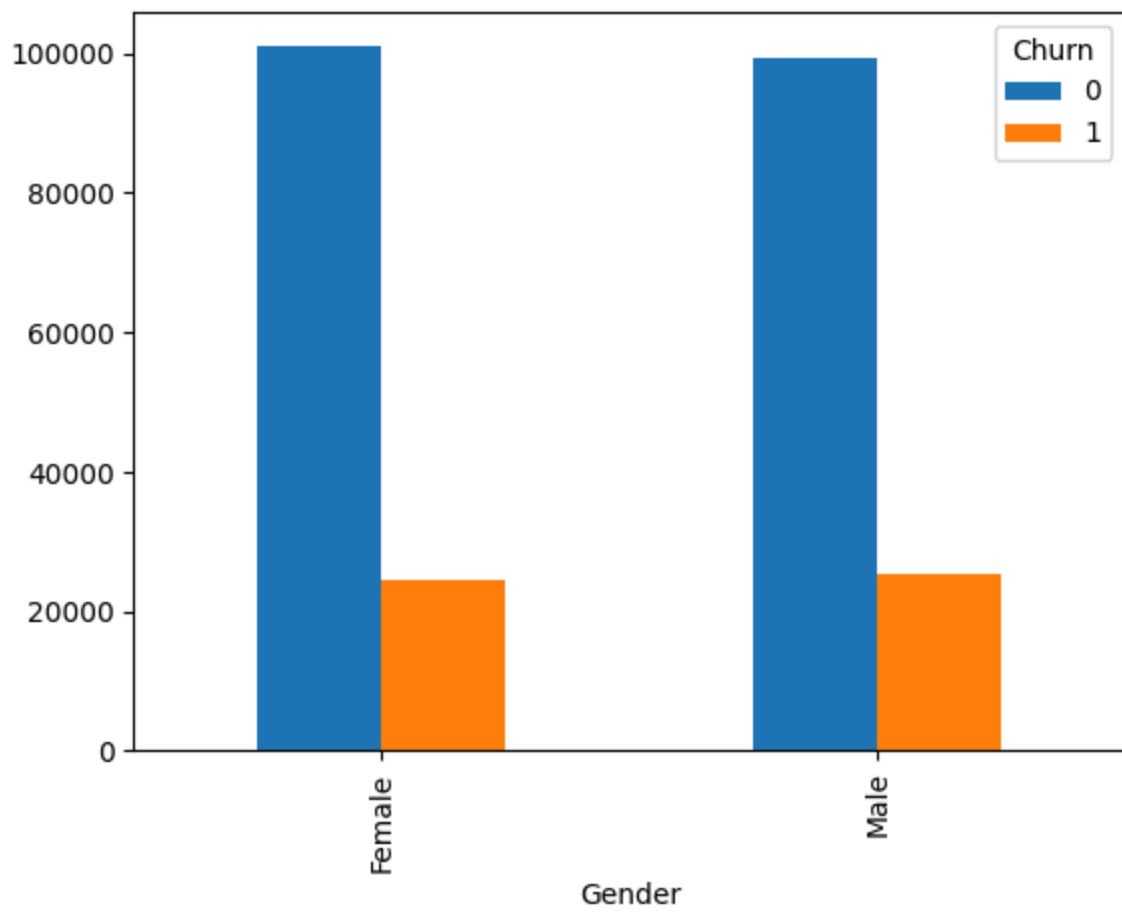


<Figure size 640x480 with 0 Axes>

In [31]: *#Churn Analysis*

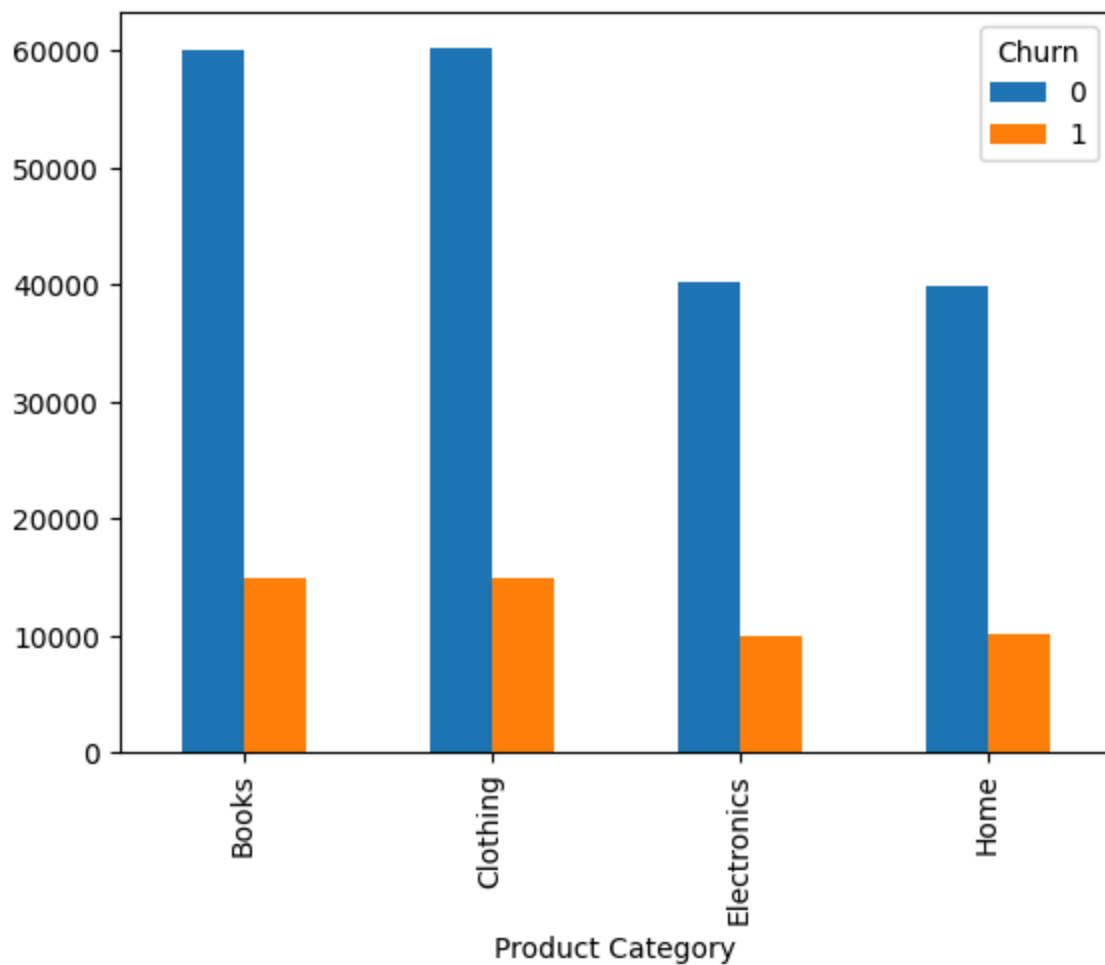
*#Churn by Gender*

```
pd.crosstab(  
    df['Gender'],  
    df['Churn']  
)  
.plot(kind='bar')  
plt.savefig("churn_gender.png",dpi=300,bbox_inches="tight")
```



In [32]: *#Churn by Product Category*

```
pd.crosstab(  
    df['Product Category'],  
    df['Churn']  
) .plot(kind='bar')  
plt.savefig("churn_product.png", dpi=300, bbox_inches="tight")
```



```
In [33]: print("Key Observations:\n")

print("1. Books and Clothing are the most popular product categories, each rec
print("2. Books and Clothing generate the highest revenue, contributing nearly
print("3. The customer base is evenly distributed by gender, with Female custo
print("4. Customer churn rate is approximately 20%, with around 50,000 custome
print("5. Credit Card is the most preferred payment method among customers, fo
print("6. Monthly revenue trends remain relatively stable throughout the analy
print("7. Monthly active customer counts also remain stable, suggesting steady
print("8. RFM segmentation identified four key customer groups: Regular Custom
print("9. Regular Customers form the largest customer segment, while Loyal Cus
print("10. Champions represent the most valuable customers due to their high p
print("11. At-Risk Customers represent a substantial portion of the customer b
```

```

print("12. Churn patterns appear relatively consistent across genders and product categories.")
print("    Insight: Customer retention strategies should focus on overall customer engagement.")

# Insights

print("1. Books and Clothing are the most popular product categories, each accounting for 25% of total revenue.")
print("    Insight: Customers show the highest interest in Books and Clothing products.")

print("2. Books and Clothing generate the highest total revenue, contributing 50% of the total.")
print("    Insight: These categories are the primary revenue drivers and have the highest profit margins.")

print("3. Female customers represent 50.2% of the customer base, while Male customers represent 49.8%.")
print("    Insight: The customer base is well-balanced across genders, indicating a broad market appeal.")

print("4. Approximately 20% of customers have churned, while 80% remain active.")
print("    Insight: Although retention is relatively strong, there is significant room for improvement in customer loyalty.")

print("5. Credit Card is the most preferred payment method, followed by PayPal and Cash.")
print("    Insight: Customers prefer digital payment options over Cash and Cryptocurrency.")

print("6. Monthly revenue remains relatively stable throughout the analysis period, with a slight upward trend.")
print("    Insight: Business performance is consistent with no major seasonal fluctuations.")

print("7. Monthly active customer counts remain stable over time.")
print("    Insight: Customer engagement levels are steady and predictable.")

print("8. RFM segmentation identified Regular Customers as the largest customer segment, followed by Loyal Customers.")
print("    Insight: Most customers purchase occasionally and can be nurtured into more frequent buyers.")

print("9. Loyal Customers form a significant portion of the customer base.")
print("    Insight: Existing customer loyalty provides a strong foundation for future growth.")

print("10. Champions are the smallest segment but represent the highest-value customers.")
print("    Insight: Retaining these customers is critical because they contribute disproportionately to revenue.")

print("11. At-Risk Customers represent a substantial segment.")
print("    Insight: Targeted retention campaigns could prevent future revenue loss.")

print("12. Churn patterns are similar across genders and product categories.")
print("    Insight: Customer retention strategies should focus on overall customer engagement.")

print("Recommendations:")

print("1. Introduce a loyalty rewards program for Champions and Loyal Customers to encourage repeat purchases.")
print("2. Develop targeted re-engagement campaigns for At-Risk Customers using personalized offers.")
print("3. Increase marketing investments in Books and Clothing categories, as they are the primary revenue drivers.")
print("4. Use customer segmentation insights to create personalized marketing campaigns for each segment.")

```



```
print("5. Optimize and promote digital payment methods, especially Credit Card
```

## Key Observations:

1. Books and Clothing are the most popular product categories, each recording approximately 75,000 purchases, while Electronics and Home categories have comparatively lower purchase volumes.
2. Books and Clothing generate the highest revenue, contributing nearly ₹205 million each, making them the strongest-performing product categories.
3. The customer base is evenly distributed by gender, with Female customers accounting for 50.2% and Male customers accounting for 49.8%, indicating balanced customer participation.
4. Customer churn rate is approximately 20%, with around 50,000 customers churned and 200,000 customers remaining active, highlighting the importance of retention strategies.
5. Credit Card is the most preferred payment method among customers, followed by PayPal, Cash, and Crypto.
6. Monthly revenue trends remain relatively stable throughout the analysis period, indicating consistent business performance with no significant seasonal fluctuations.
7. Monthly active customer counts also remain stable, suggesting steady customer engagement over time.
8. RFM segmentation identified four key customer groups: Regular Customers, Loyal Customers, At-Risk Customers, and Champions.
9. Regular Customers form the largest customer segment, while Loyal Customers contribute significantly to overall business performance.
10. Champions represent the most valuable customers due to their high purchase frequency and spending behavior, despite being the smallest segment.
11. At-Risk Customers represent a substantial portion of the customer base and should be targeted through personalized retention campaigns.
12. Churn patterns appear relatively consistent across genders and product categories, suggesting that retention efforts should focus on overall customer engagement and experience.

## Key Insights from Customer Behavior Analysis :

1. Books and Clothing are the most popular product categories, each accounting for approximately 75,000 purchases.

Insight: Customers show the highest interest in Books and Clothing products compared to Electronics and Home products.

2. Books and Clothing generate the highest total revenue, contributing nearly ₹205 million each.

Insight: These categories are the primary revenue drivers and have the greatest impact on overall business performance.

3. Female customers represent 50.2% of the customer base, while Male customers represent 49.8%.

Insight: The customer base is well-balanced across genders, indicating broad market appeal.

4. Approximately 20% of customers have churned, while 80% remain active.

Insight: Although retention is relatively strong, there is significant opportunity to reduce customer churn.

5. Credit Card is the most preferred payment method, followed by PayPal.

Insight: Customers prefer digital payment options over Cash and Crypto transactions.

6. Monthly revenue remains relatively stable throughout the analysis period.

Insight: Business performance is consistent with no major seasonal fluctuations.

7. Monthly active customer counts remain stable over time.

Insight: Customer engagement levels are steady and predictable.

8. RFM segmentation identified Regular Customers as the largest customer segment.

Insight: Most customers purchase occasionally and can be nurtured into more loyal segments.

9. Loyal Customers form a significant portion of the customer base.

Insight: Existing customer loyalty provides a strong foundation for future growth.

10. Champions are the smallest segment but represent the highest-value customers.

Insight: Retaining these customers is critical because they contribute disproportionately to revenue.

11. At-Risk Customers represent a substantial segment.

Insight: Targeted retention campaigns could prevent future revenue loss.

12. Churn patterns are similar across genders and product categories.

Insight: Customer retention strategies should focus on overall customer experience rather than a specific demographic.

#### Recommendations:

1. Introduce a loyalty rewards program for Champions and Loyal Customers to improve retention and increase lifetime value.

2. Develop targeted re-engagement campaigns for At-Risk Customers using personalized offers and discounts.

3. Increase marketing investments in Books and Clothing categories, as they generate the highest revenue and customer interest.

4. Use customer segmentation insights to create personalized marketing campaigns for different customer groups.

5. Optimize and promote digital payment methods, especially Credit Cards and PayPal, to enhance customer convenience and transaction efficiency.